Cross-Species Transfer Learning with DeepSEA: From Human to Mouse

Eric Xu **Alexander Speltzer**

Abstract

We test whether the convolutional features a DeepSEA model learns on the human genome transfer to mouse. We reuse the three convolutional layers of human DeepSEA, attach a fresh multi-task head, and fine-tune on roughly one million 1000-bp windows of mouse (mm10) ENCODE data spanning 133 chromatin features. Starting from human weights lowers validation loss faster and reaches a slightly higher held-out AUROC (0.877) than training the same network from scratch (0.865). The gap is small but consistent: transfer wins on 115 of 133 features and helps most on the rarest targets, yet it shows up mainly as faster convergence rather than a higher accuracy ceiling. This suggests that the early sequence-motif detectors learned on human are partly shared across species, and that warm-starting is most useful when mouse data or compute is limited.

1 Introduction

DeepSEA is a convolutional neural network that predicts chromatin features such as transcription factor (TF) binding, DNase I hypersensitivity, and histone marks directly from DNA sequence [Zhou and Troyanskaya, 2015]. Its first convolutional layers act as motif detectors, learning short sequence patterns that resemble TF binding motifs. Because the regulatory code is built from such motifs, a natural question is whether these learned detectors are specific to the human genome or whether they capture patterns shared across species.

We study this through transfer learning from human to mouse. If the early layers of DeepSEA encode motifs that are conserved across species, then initializing a mouse model with human-trained convolutional weights should help, both by speeding up training and by improving accuracy, relative to training the same network from random initialization. A positive result would be evidence that DeepSEA learns common motifs rather than human-specific artifacts.

Related work. Our work builds directly on DeepSEA [Zhou and Troyanskaya, 2015], which introduced the architecture, the 1000-bp input window, and the multi-task formulation that predicts hundreds of chromatin features at once. DeepSEA was trained and evaluated only on the human genome. We keep its convolutional design and training setup and ask whether the features it learns extend to a second species.

2 Methods

Architecture. We split DeepSEA into a *trunk* and a *head*. The trunk is the three convolutional blocks of the original model: convolutions with 320, 480, and 960 kernels of width 8, each followed by a ReLU, with max-pooling (width 4) after the first two blocks and dropout throughout. The trunk maps a one-hot encoded 4×1000 input to a $960 \times L$ feature map. The head flattens this map and applies a linear layer with a sigmoid output per feature. The head is sized to the mouse feature set, so it is always trained from scratch. Only the trunk can carry transferred knowledge, because the human and mouse label sets do not line up one to one.

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Table 1: Held-out (chr19) performance, median over 132 evaluable features.

Condition	Trunk init	Fine-tuned	Median AUROC	Median AUPRC
Baseline	Human	No	0.496	0.018
Scratch	Random	Yes	0.865	0.230
Transfer	Human	Yes	**0.877**	**0.250**

Transfer. Human weights come from the published DeepSEA checkpoint. The original weights are stored as 2D convolutions with a singleton dimension, so we squeeze them to 1D convolution kernels and copy the three convolutional layers into our trunk by shape-matched order. We then compare two conditions that share all other settings: **Transfer**, where the trunk starts from human weights, and **Scratch**, where the trunk starts from random initialization. As a control we also report a **Baseline**, which uses the human trunk with an untrained head and is evaluated with no fine-tuning.

Data. We built a DeepSEA-style multi-task dataset for mouse (mm10) from ENCODE. We queried released narrowPeak files for TF ChIP-seq, histone ChIP-seq, and DNase-seq across five biosamples: liver, heart, forebrain, CH12.LX, and MEL. A feature is a (biosample, target) pair, for example `liver|CTCF`, so the same target in several cell types counts as several features. We binned the genome into 200-bp bins and labeled a bin positive for a feature when more than half of the bin overlaps a peak of that feature. Each retained bin was expanded to a 1000-bp window centered on it and one-hot encoded over $\{A, C, G, T\}$, with unknown bases mapped to an all-zero column. We subsampled to 10^6 windows over 133 features.

Training and evaluation. We split by chromosome to avoid leakage: chr19 is the test set and chr18 is the validation set, both held out entirely, leaving about 945,000 training windows. We train with AdamW and binary cross-entropy with logits, using a per-feature positive weight equal to the negative-to-positive ratio (clipped to $[1, 100]$) to handle class imbalance. We use a learning rate of 10^{-3} for the head, batch size 256, a plateau learning-rate scheduler, and early stopping on validation loss. We report per-feature AUROC and AUPRC on chr19, summarized by their median across the 132 features that have both positive and negative labels in the test chromosome. All models were trained on Google Colab using an NVIDIA L4 GPU.

3 Results

Transfer trains faster and scores slightly higher. Table 1 reports the held-out metrics for the three conditions. We score every feature with AUROC, the area under the ROC curve, where 0.5 is chance and 1.0 is perfect, and with AUPRC, the area under the precision-recall curve, which is more informative than AUROC when positives are rare. The transfer model reaches a median AUROC of 0.877, just above the from-scratch model at 0.865, and a higher median AUPRC (0.250 versus 0.230). The zero-shot baseline sits at chance (AUROC 0.496), as expected, since its head is never trained. Figure 1 plots the training and validation loss (binary cross-entropy, y axis) against training epoch (x axis); lower is better. The difference in training dynamics is clearer than the difference in final score: starting from human weights, the validation loss drops faster and reaches its minimum of 0.7277 at epoch 5, while the from-scratch model needs until epoch 12 to reach a slightly worse minimum of 0.7403. The human-initialized model also starts from a lower training loss in the first epoch, indicating that the transferred filters are already partly useful before any mouse fine-tuning.

The warm start is worth several epochs of training. Tracking the held-out metric directly tells the same story more sharply. Figure 2 plots the validation median AUROC (y axis) at each training epoch (x axis). The warm-started model is already at 0.846 after a single epoch, which is roughly where the from-scratch model arrives only after five, and it plateaus near 0.885 by epoch 5 while the from-scratch model is still climbing at epoch 10 (0.870). The transferred filters give the optimizer a large head start, so the human trunk's most reliable benefit is reaching a good model in far less training, not a higher final ceiling.

Per-feature behavior matches the original DeepSEA. Figure 3 is a histogram of the transfer model's per-feature AUROC: the x axis is AUROC and the y axis counts how many features fall in

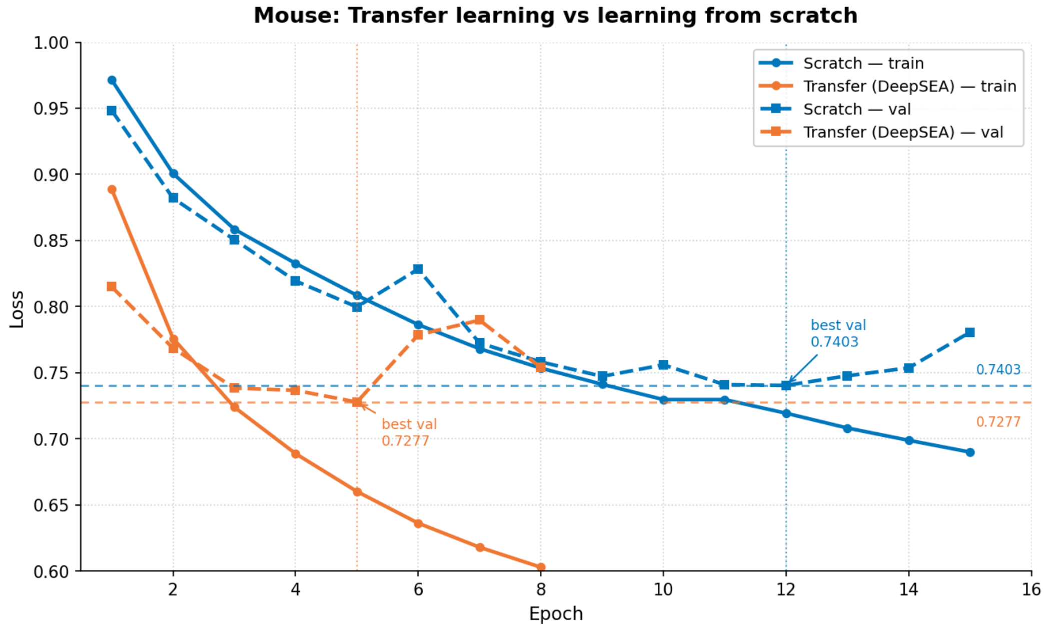

Figure 1: Training and validation loss for transfer versus from-scratch. The human-initialized model (orange) decreases loss faster and reaches a lower best validation loss (0.7277 at epoch 5) than the random-initialized model (blue, 0.7403 at epoch 12). Dashed horizontal lines mark each best validation loss.

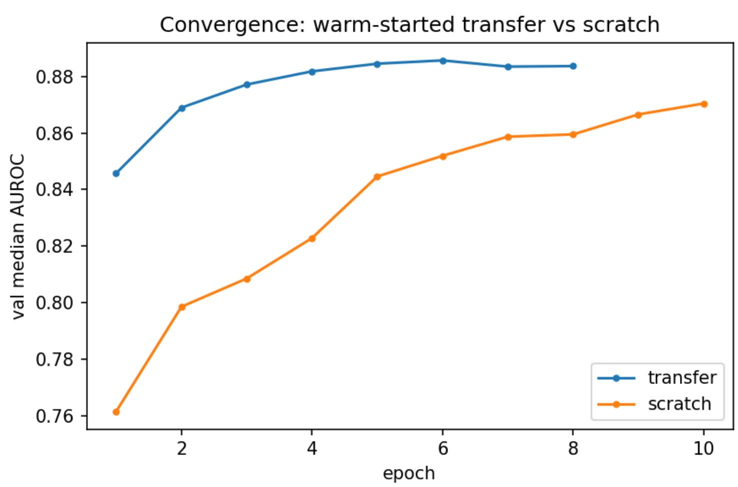

Figure 2: Validation median AUROC per epoch for transfer (blue) versus from-scratch (orange). Transfer after one epoch matches from-scratch after about five and saturates near 0.885 within five epochs; from-scratch is still rising at epoch 10. Early stopping halts the transfer run at epoch 8.

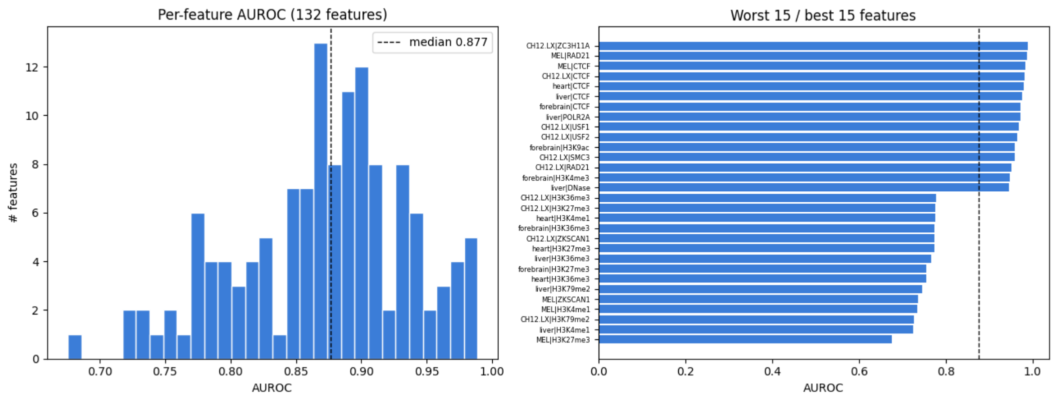

Figure 3: Per-feature AUROC of the transfer model on chr19. Left: distribution over 132 features, median 0.877. Right: the 15 best and 15 worst features. TF and structural targets (CTCF, RAD21, USF, POLR2A) score highest; broad histone marks score lowest.

each bin. The model predicts TF and structural targets best: CTCF (across all cell types), RAD21, SMC3, USF1/2, POLR2A, and ZC3H11A all exceed an AUROC of 0.95. The hardest features are repressive and broad histone marks such as H3K27me3, H3K4me1, H3K36me3, and H3K79me2, which fall to 0.68 to 0.78. This ordering, sharp TF and accessibility signals easy and broad histone marks hard, is the same pattern reported for human DeepSEA [Zhou and Troyanskaya, 2015].

The transfer gain is small but consistent across features. Because the median gap is only 0.012, we check whether it reflects a broad tendency or a few lucky features. In Figure 4 each point is one feature, with its from-scratch AUROC on the x axis and its transfer AUROC on the y axis; points above the dashed diagonal are features that transfer improves. Transfer wins on 115 of 133 features, with a median per-feature gain of about 0.007, and the points sit just above the diagonal across all three assay types. The one feature where transfer loses badly (AUROC 0.67 versus 0.97) is the rarest TF target, which has too few positives in chr19 for a stable AUROC. That so many independent features move the same way argues the advantage is a real if modest effect, not noise in the median.

Transfer helps every assay type; the baseline is at chance. Figure 5 groups the features by assay type along the x axis and plots each condition's median AUROC (y axis) as a bar. Transfer is at or above from-scratch for TF (0.89 versus 0.88), histone (0.82 versus 0.81), and DNase (0.93 versus 0.92), while the zero-shot baseline sits at chance (about 0.49) for all three, confirming the untrained head learns nothing on its own. The ordering, DNase and TF easy and broad histone marks hard, matches both the per-feature plot and human DeepSEA.

The gain concentrates on rarer features. In Figure 6 the y axis is each feature's transfer gain (its transfer AUROC minus its from-scratch AUROC, so positive means transfer helps) and the x axis is the feature's positive rate in the test chromosome on a log scale, with rarer features to the left. The largest positive gains fall on rarer targets (positive rate near 10^{-3} to 10^{-2}), while common features are essentially tied. This is the expected signature of a useful prior: the human filters help most where mouse positives are scarce and a from-scratch model has little signal to learn from, and they help little once a feature has enough positives to be learned directly.

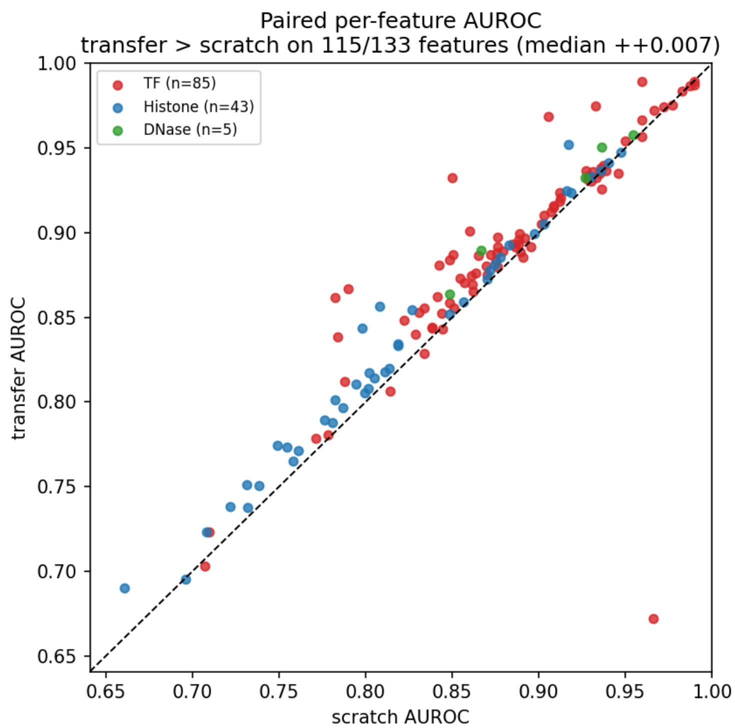

Figure 4: Per-feature AUROC, transfer (vertical) versus from-scratch (horizontal); points above the dashed diagonal favor transfer. Transfer wins on 115 of 133 features (median gain $+0.007$). Colors mark assay type. The single far-below-diagonal point is the rarest TF feature, where AUROC is unstable.

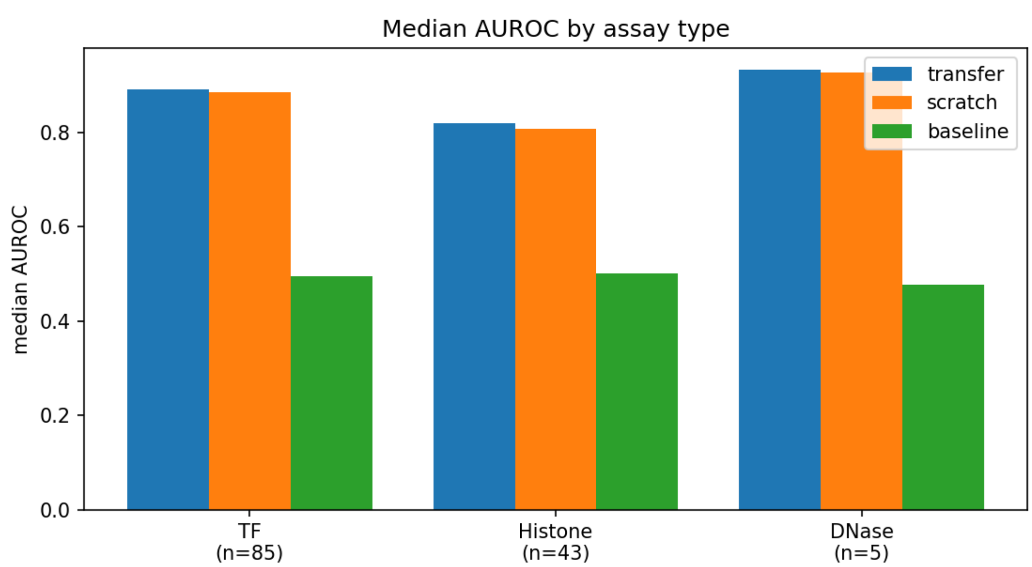

Figure 5: Median AUROC by assay type for the three conditions. Transfer (blue) edges out from-scratch (orange) in every assay class; the baseline (green) is at chance everywhere. DNase and TF are easiest, broad histone marks hardest.

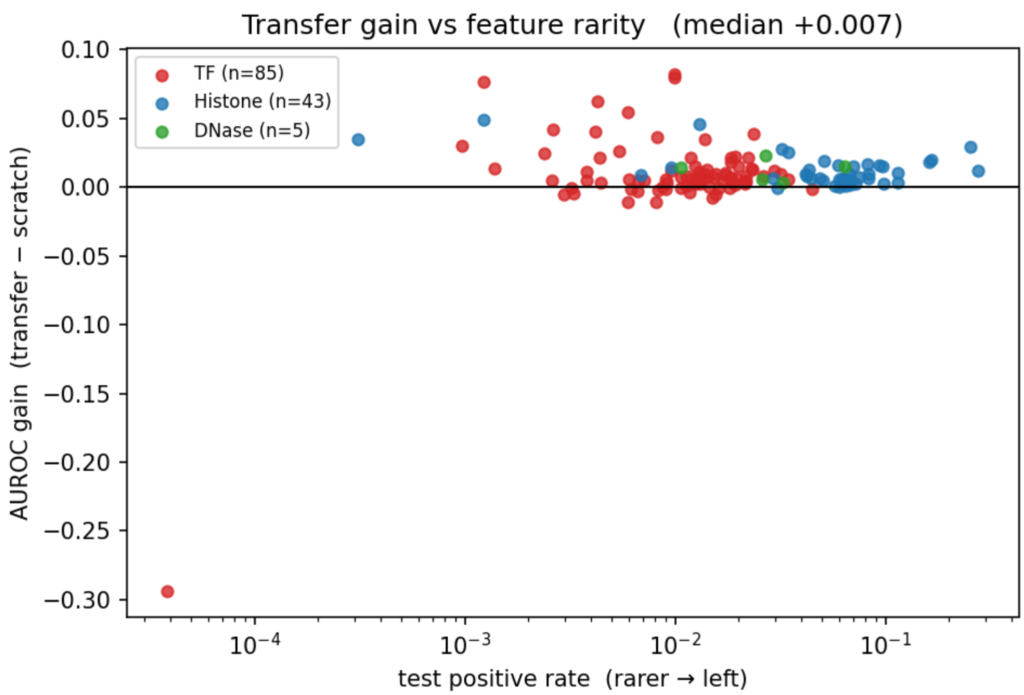

Figure 6: Per-feature AUROC gain (transfer minus from-scratch) versus test positive rate (rarer to the left). Gains are largest for rarer features and shrink toward zero as features become common. The lone large negative is the rarest TF target from Figure 4.

4 Discussion

Transfer learning from human to mouse works, but mainly as an efficiency gain. The human-initialized trunk reaches a good model in a fraction of the training, wins on 115 of 133 features, and helps most on the rarest targets, where labeled positives are scarce; yet the final-accuracy gap is small (0.877 versus 0.865 median AUROC). This fits the idea that the early convolutional filters encode sequence motifs shared across the two species, so a warm-started model does not have to relearn the basic grammar and can get good chromatin-mark predictions for a new species or assay set with less data and compute. The flip side is that the gap is not a higher ceiling: given enough mouse data a from-scratch model could likely learn the same motifs and catch up, so transfer is most worth it when data or compute is the binding constraint. Our absolute scores are below human DeepSEA (median AUROC about 0.958 for TF, 0.923 for DNase, 0.856 for histone marks [Zhou and Troyanskaya, 2015]), consistent with our smaller training budget and feature set rather than the method. We leave fuller training and more distant species for future work.

References

Jian Zhou and Olga G. Troyanskaya. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature Methods*, 12(10):931–934, 2015.